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iteration	a	b	c
0	4	4	6
1	3	3	8
2	2	2	10
3	1	1	12
4	0	0	14

Table 7.2: Series of meetings between two types of chameleons equal in number.

In each meeting, a and b each decreases by 1, and c increases by 2. The solution can be expressed as an iterative algorithm.

monochromatize(a, b, c)

-- **inputs: a=A, b=B, c=C, a=b**

-- **outputs: a = b = 0, c = A+B+C**

while a > 0

a, b, c := a-1, b-1, c+2

The algorithm is depicted in the flowchart of Figure 7.7.

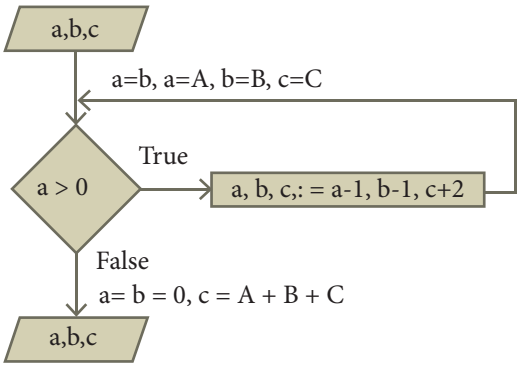

Figure 7.7: Algorithm monochromatize

7.3 Decomposition

Problem decomposition is one of the elementary problem-solving techniques. It involves breaking down a problem into smaller and more manageable problems, and combining the solutions of the smaller problems to solve the original problem. Often, problems have structure. We can exploit the structure of the problem and break it into smaller problems. Then, the smaller problems can be further broken until they become sufficiently small to be solved by other simpler means. Their solutions are then combined together to construct a solution to the original problem.

7.3.1 Refinement

After decomposing a problem into smaller subproblems, the next step is either to refine the subproblem or to abstract the subproblem.

1. Each subproblem can be expanded into more detailed steps. Each step can be further expanded to still finer steps, and so on. This is known as refinement.

2. We can also abstract the subproblem. We specify each subproblem by its input property and the input-output relation. While solving the main problem, we only need to know the specification of the subproblems. We do not need to know how the subproblems are solved.

Example 7.6. Consider a school goer's action in the morning. The action can be written as

1 Get ready for school

We can decompose this action into smaller, more manageable action steps which she takes in sequence:

1 Eat breakfast